

Spoken Tutorial – Understanding Artificial Intelligence, Machine Learning, and Deep Learning

Title: Understanding Artificial Intelligence, Machine Learning, and Deep Learning

Module	Introduction to AI/ML
Episode	1
Learning Objective	At the end of this tutorial learner will be able to 1. Understand the basic differences between Artificial Intelligence, Machine Learning, and Deep Learning.
Approx. Duration	10 minutes
Outline	• What is Artificial Intelligence (AI)? • What is Machine Learning (ML)? • ML vs Traditional Programming • What is Deep Learning? • Deep Learning in Action
Meta Tags	AI, Artificial Intelligence, ML, Machine Learning, DL, Deep Learning, Neural Networks, Computer Science, Introduction
Pre-requisite Tutorial	No prior knowledge of Artificial Intelligence or programming is required.

Script

Visual Cue	Narration
	Welcome to this Spoken Tutorial. It is about Artificial Intelligence, Machine Learning , and Deep Learning . Here, we will learn about these important computer concepts.
	After this tutorial, you will know the main differences between Artificial Intelligence, Machine Learning , and Deep Learning .
	To follow this tutorial, you do not need any special computer. You only need a computer with internet.
	This tutorial is for new learners. You do not need to know about Artificial Intelligence or programming before.
A person looking thoughtful, surrounded by various smart devices like a smartphone, smart speaker, and a laptop, with question marks floating around, simple illustration	Have you used a smart phone or app that seems to know what you want? Or a search engine that gives good results? These systems are smart because of AI .

A robot brain with gears and light bulbs, representing intelligence and computer science, simple illustration	Artificial Intelligence, or AI, is a big part of computer science. It makes machines do tasks that usually need human thinking. For example, AI helps machines understand language, see images, and solve hard problems.
A computer screen showing data points forming a pattern, with a magnifying glass over it, representing learning from data, simple illustration	Machine Learning, or ML, is a type of Artificial Intelligence. Here, computers learn from a lot of information, called data. They find patterns in the data by themselves, instead of us giving them rules.
A flow chart showing 'Pointy Ears?' -> 'Whiskers?' -> 'Cat!', representing explicit rules, simple illustration	Let's see an example. In old way of programming, you give the computer exact rules. For example, to find a cat, you tell the computer a rule. 'IF it has pointy ears AND whiskers, THEN it is a cat.'
A pile of diverse cat pictures being fed into a computer, which then outputs 'Cat Identified', representing learning from data, simple illustration	With Machine Learning, the way is different. You show the computer many pictures marked 'cat' or 'not cat'. The computer then learns by itself what makes a cat.
A simplified diagram of a multi-layered neural network with interconnected nodes, resembling brain neurons, simple illustration	Deep Learning is a more advanced part of Machine Learning. It uses special computer models called neural networks. These networks work a bit like the human brain.
A self-driving car on a road, with bounding boxes around pedestrians and other cars, and a person speaking into a smart speaker, simple illustration	Deep Learning helps computers learn very complex patterns from data. For example, self-driving cars use it to see people and other cars. Voice assistants also use deep learning to understand what you say.
	In this tutorial, we learned about AI, ML, and Deep Learning. AI is about making smart machines. ML is an AI type where machines learn from data. Deep Learning is advanced ML using neural networks.
	Think about an app you use daily. Does it use Artificial Intelligence, Machine Learning, or Deep Learning? Discuss this with your friend.
	This Spoken Tutorial is from EduPyramids Educational Services Private Limited. It is supported by SINE, IIT Bombay. Thank you for watching!